Chapter 2

REPRESENTATION OF COMMUNICATION SIGNALS AND SYSTEMS

2.1 Introduction

The process of (electronic) communication involves the generation, transmission and reception of various types of signals. The communication process becomes fairly difficult, because: a) Transmitted signals may have to travel long distances (there by undergoing severe attenuation) before they reach the destination i.e., the receiver.

b) Imperfections of the channel over which the signals have to travel

c) Interference due to other signals sharing the same channel and

d) Noise at the receiver input.

In quite a few situations, the desired signal strength at the receiver input may not be significantly stronger than the disturbance component present at that point in the communication chain. (But for the above causes, the process of communication would have been quite easy, if not trivial). In order to come up with appropriate signal processing techniques, which enable us to extract the desired signal from a distorted and noisy version of the transmitted signal, we must clearly understand the nature and properties of the desired and undesired signals present at various stages of a communication system. In this lesson, we begin our study of this aspect of communication theory.

Signals physically exist in the time domain and are usually expressed as a function of the time parameter. Because of this feature, it is not too difficult, at least in the majority of the situations of interest to us, to visualize the signal behavior in the Time Domain. In fact, it may even be possible to view the signals on an oscilloscope. But equally important is the characterization of the signals in the Frequency Domain or Spectral Domain. That is, we characterize the signal in terms of its various frequency components (or its spectrum). Fourier analysis (Fourier Series and Fourier Transform) helps us in arriving at the spectral description of the pertinent signals.

2.2 Signals and Their Classifications

A signal is defined as a single-valued function of time that conveys information. Consequently, for every instant of time there is a unique value of the function. This value may be a real number

in which case we have a real-valued signal, or it may be a complex number, in which variable (namely time) is real-valued. For a given situation, the most useful method of signal representation hinges on the particular type of signal being considered. Depending on the feature of interest, we may identify four different methods of dividing signals into two classes.

2.2.1 Periodic Signals and Non-Periodic Signals

A periodic signal (t) is a function that satisfies the condition

$(t) = (t + T_o)$ … … … … … … … … … … … … . . (2.1)

For all t, where t denotes time and $(T_o \neq 0)$ is a constant. The smallest value of T_o that satisfies this condition is called the period of (t). Accordingly, the period T_o defines the duration of one complete cycle of (t). Any signal for which there is no value of T_o to satisfy the condition of equation (2.1) is called non-periodic or aperiodic signals.

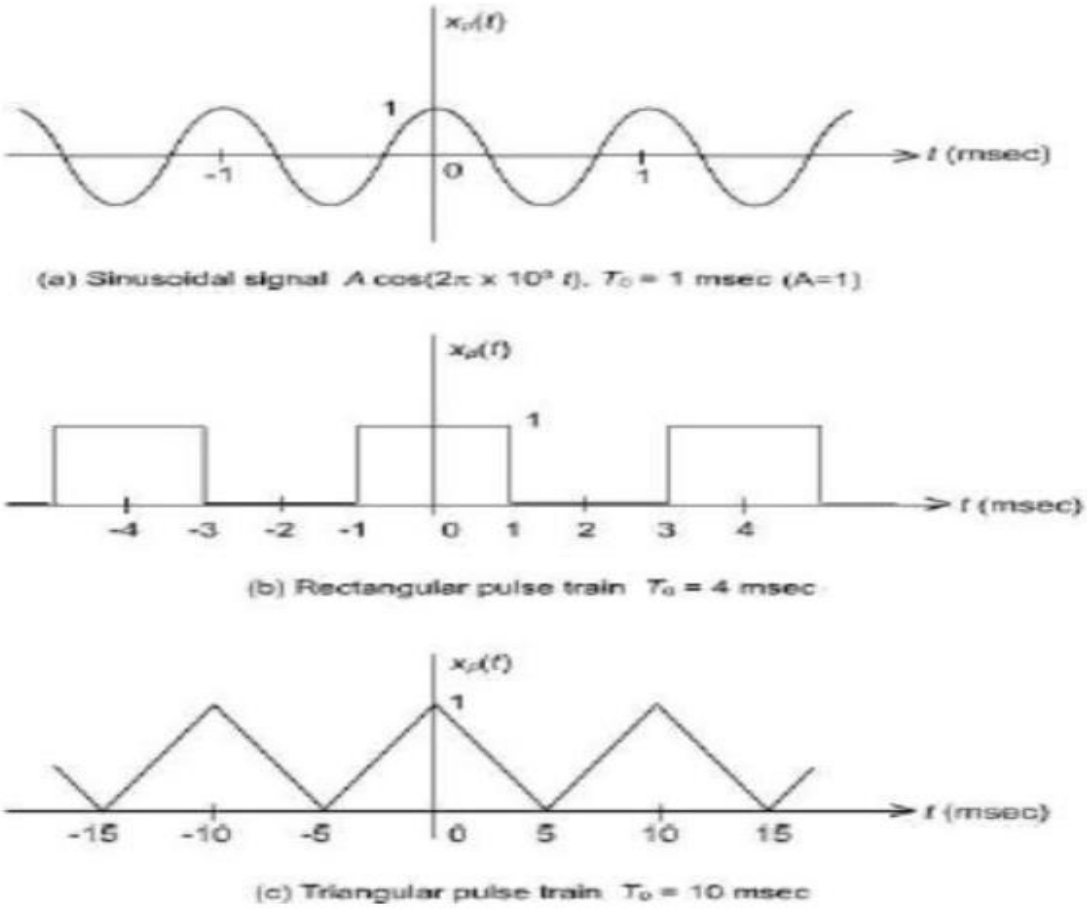


Fig. 2.1 shows some examples of periodic signals

2.2.2 Deterministic Signals and Random Signals

A deterministic signal is a signal about which there is no uncertainty with respect to its value at any time. Accordingly, we find that deterministic signals may be modeled as completely specified functions of time. On the other hand, a random signal is a signal about which there is uncertainty before its actual occurrence. Such a signal may be viewed as belonging to an ensemble of signals, with each signal in the collection having a different wave form. Moreover, each signal within the ensemble has a certain probability of occurrence. e.g. (t) = $Ac(\pi t + \theta)$, if A and θ are known constants or if there is no uncertainty in A or θ, I can predict value of $x(t)$ at any given time t, $x(t)$ is deterministic signal. But if either A or θ are known with certainty, (t) becomes random signal. *i.e.* if the value of A = {-1, 1} are determined based on an experiment (coin toss) and θ is a known constant, then the value of $x(t)$ at a given time t is determined by the outcome of the coin toss, $X(t)$ is random signal.

2.2.3 Energy Signals and Power Signals

In communication systems, a signal may represent a voltage or a current. Consider a voltage (t) developed across a resistor R, producing a current $i(t)$. The instantaneous power dissipated in this resistor is defined by

$$p = v^2(t)/R \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2.2)$$

Or, equivalently,

$$p = Ri^2(t) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2.3)$$

In both cases, the instantaneous power p is proportional to the squared amplitude of the signal. Furthermore, for a resistance R of 1Ω, we see that equations (2.2) and (2.3) take on the same mathematical form. Accordingly, in signal analysis it is customary to work with a 1Ω resistor, so that, regardless of whether a given signal (t) represents a voltage or a current, we may express the instantaneous power associated with the signal as

$$p = g^2(t) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2.4)$$

Based on this convention, we define the total energy of a signal (t) as

$$E = \lim_{T \to x} \int_{-T}^{T} g^2(t)dt$$
$$= \int_{-\infty}^{\infty} g^2(t)dt \dots\dots\dots\dots (2.5)$$

and its average power as

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} g^2(t)dt \dots\dots\dots\dots\dots\dots (2.6)$$

We say that the signal (t) is an energy signal if and only if the total energy of the signal satisfies the condition

$$0 < E < \infty$$

We say that the signal (t) is a power signal if and only if the average power of the signal satisfies the condition

$$0 < P < \infty$$

The energy and power classifications of signals are mutually exclusive. In particular, an energy signal has zero average power, where as a power signal has infinite energy. Also it is of interest to note that, usually periodic signals and random signals are power signals, whereas signals that are both deterministic and non-periodic are energy signals.

2.2.4 Analog Signals and Digital Signals

An analog signal is a signal with amplitude (*i.e.*, value of signal at some fixed time) that varies continuously for all time; that is, both amplitude and time are continuous over their respective intervals. Analog signals arise when a physical waveform such as an acoustic wave or a light wave is converted into an electrical signal. The conversion is effected by means of a transducer; examples include the microphone, which converts sound pressure variations into corresponding voltage or current variations, and the photo-detector cell, which does the same for light intensity variations. On the other hand, a discrete-time signal is defined only at discrete instants of time. Thus, in this case, the independent variable takes on only discrete values, which are usually uniformly spaced. Consequently, discrete time signals are described as sequences of samples that may take on a continuum of values. When each sample of a discrete-time signal is quantized (i.e., it is only allowed to take on a finite set of discrete values) and then coded, the resulting signal is referred to as a digital signal. The output of a digital computer is an example of a digital

signal. Naturally, an analog signal may be converted into digital form by sampling in time, then quantizing and coding.

2.2.5 Periodic Signals and Fourier series

Just as we can express any signal as the sum of a series of shifted and scaled impulses, so too we can express any signal as the sum of a series of (shifted and scaled) sinusoids at different frequencies. This is called the Fourier expansion of the signal. The idea of a Fourier series is that any (reasonable) function, (x), that is periodic can be decomposed into contributions from $s(nx)$ and $cos(nx)$. Let (x) be a periodic signal with period T_o. Then $f_o = 1/T_o$ is called the fundamental frequency and nf_o is called the n_{th} harmonic, where n is an integer (for $n = 0$, we have the DC component). The general Fourier series may be written as:

$$f(x) = \frac{a_0}{2} + a_1\cos(x) + a_2\cos(2x) + a_3\cos(3x) + \cdots + a_n\cos(nx) + b_1\sin(x) + b_2\sin(2x) + b_3\sin(3x) + \cdots + b_n\sin(nx)$$

Note:

1. $Cos(nx)$ and $sin(nx)$ are periodic on the interval 2π for any integer n.
2. The a_n and b_n coefficients measure the strength of contribution from each harmonic

Coefficient a_0, a_n or b_n respectively, that is:

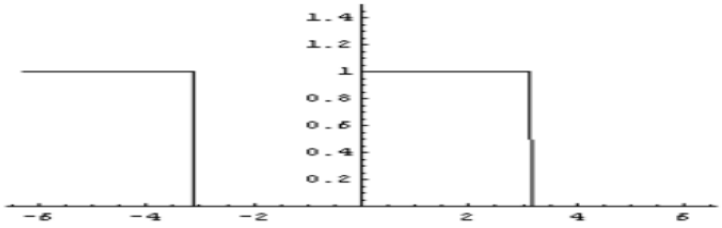
$$a_0 = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)\cos(nx)dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)\sin(nx)dx$$

Example:- Consider the square wave:

$$f(x) = 1, \quad 0 \le x < \pi$$
$$= 0, \quad -\pi \le x < 0$$



The coefficients in the expansion can be determined from the formulae given above. a_0 is determined by:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(x)dx = \frac{1}{\pi}\int_{0}^{+\pi} 1dx = \frac{1}{\pi} * \pi = 1$$

where the restriction of the integral to the region $0 \rightarrow \pi$ is simply because $f(x)$ is zero in the region $-\pi \rightarrow 0$. Similarly:

$$a_n = \frac{1}{\pi}\int_{0}^{+\pi} 1 * \cos(nx)\, dx = 0$$

$$a_n = 0 \quad for\ all\ n$$

For the b coefficients we have:

$$b_n = \frac{1}{\pi}\int_{0}^{+\pi} 1 * \sin(nx)\, dx$$

$$= \frac{1}{\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{1}{n\pi}(1 - \cos(n\pi))$$

but,

$$\cos(n\pi) = +1 \quad for\ n - even$$

$$= -1 \quad for\ n - odd$$

So,

$$b_n = 0 \quad n - even$$

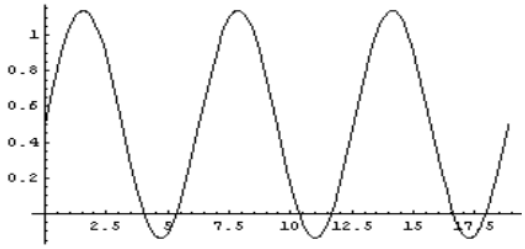
$$= \frac{2}{n\pi} \quad n - odd$$

Having determined all of the coefficients we can write the series for $f(x)$ as:

$$f(x) = \frac{1}{2} + \frac{2}{\pi}\left(\sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \cdots\right)$$

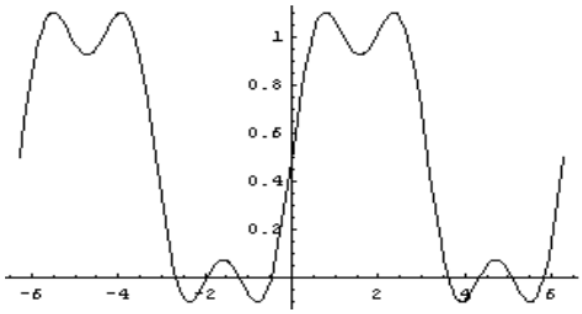
The sum continues to an infinite number of terms. We can see how it converges to the square wave by plotting the truncated sum containing a finite number of terms. Let's call the sum containing n -trigonometric terms (x) then $f_0\ (x)$ = 1/2 is just the average value of the square wave.

$f_1(x) = \frac{1}{2} + \frac{2}{\pi}\sin(x)$ is plotted below

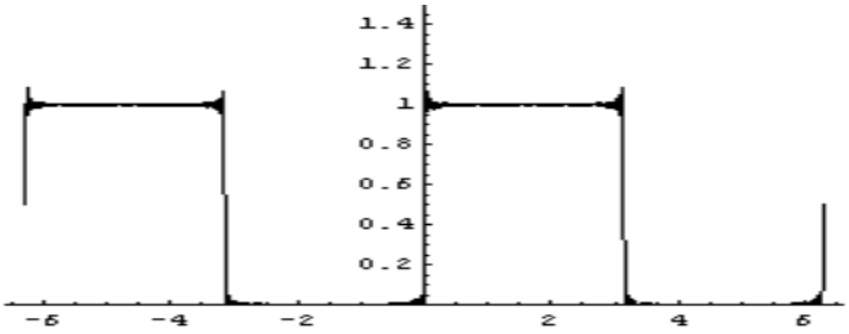


which is the "best" approximation that can be made using just a constant and a sine wave - not great.

$f_2(x) = \frac{1}{2} + \frac{2}{\pi}\sin(x) + \frac{2}{3\pi}\sin(3x)$ looks like this



Which is beginning to look more like a square well. The weight in each contribution is falling andwith each additional term the fine detail of the square wave is being refined. Including 50 terms, $f_{50}(x)$, we get



2.4.3 Fourier Transform

Like periodic signals, aperiodic signals also can be represented in the frequency domain. However, unlike the discrete spectrum of the periodic case, we have a continuous spectrum for the aperiodic case; that is, the frequency components constituting a given signal $x(t)$ lie in a continuous range (or ranges), and quite

often this range could be (-∞, ∞). Fourier series expresses (t) as a sum over a discrete set of frequencies. Its counterpart for the aperiodic case is an integral relationship given by:

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

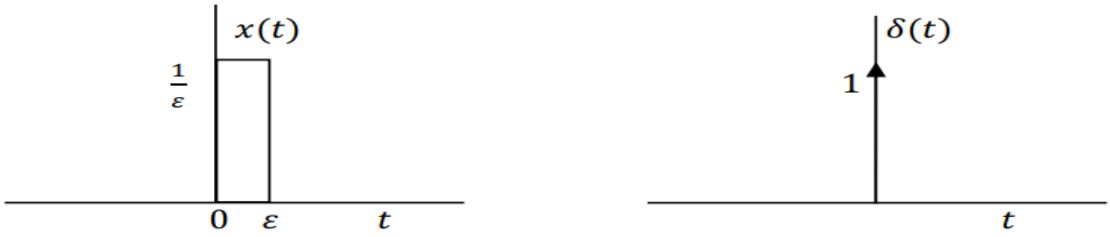
Where (f) is the Fourier transform of (t).

Eq.1 is called the synthesis relation or Inverse Fourier Transform (IFT) relation. Quite often, we know (t) and would want $X(f)$. The companion relation to Eq.1 is

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

2.5 Unit Impulse function

Let's consider a rectangular pulse (t) of width ε and amplitude $1/\varepsilon$.



Therefore, we can characterize $\delta(t)$ as a function with:

a) Amplitude $= \begin{Bmatrix} \infty; t = 0 \\ 0; t \neq 0 \end{Bmatrix}$

b) Area $= \int_0^{\varepsilon} \delta(t)dt = 1$

c) Time scale, $\delta(at) = \delta(t)/|a|$

d) Multiplication by a time function

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$$

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)\, dt = x(t_0)$$

e) Relation to step function $u(t)$

$$\delta(t) = d\{u(t)\}/dt$$

2.6 Convolution and Output of Linear Time-Invariant (LTI) System

2.6.1. Convolution

Convolution: - is an integral that expresses the amount of overlap of one function when it is shifted over another fornication.

The *convolution* of two functions (t) and (t) is the function

$$z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(t-\tau)y(\tau)\,d\tau$$

The following are some important properties of the convolution:

1. $f1 * f2 = f2 * f1$
2. $f1 *(f2 * f3) = (f1 * f2) * f3$
3. $f1 *(f2 + f3) = f1 * f2 + f1 * f3$

These results show that $f1$, $f2$, $f3$ obey the *commutative*, *associative*, and *distributive laws* of algebra with respect to the operation of convolution.

2.7.2 Measures of Bandwidth

What is network bandwidth?

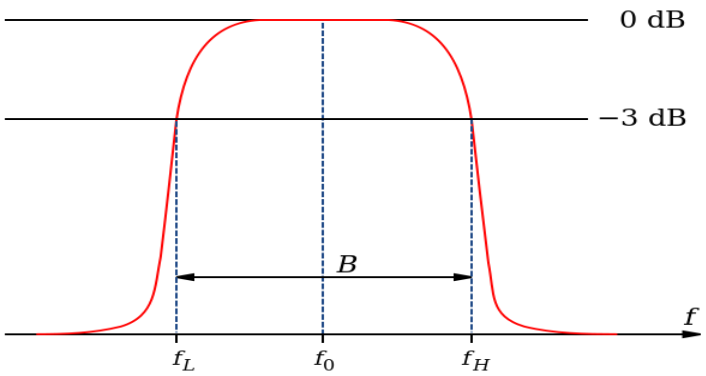
Network bandwidth is a measurement indicating the maximum capacity of a wired or wireless communications link to transmit data over a network connection in a given amount of time. Typically, bandwidth is represented in the number of bits, kilobits, megabits or gigabits that can be transmitted in 1 second. Synonymous with capacity, bandwidth describes data transfer rate. Bandwidth is not a measure of network speed -- a common misconception.

How does bandwidth work?

The more bandwidth a data connection has, the more data it can send and receive at one time. In concept, bandwidth can be compared to the volume of water that can flow through a pipe. The wider the pipe's diameter, the more water can flow through it at one time. Bandwidth works on the same principle. The higher the capacity of the communication link, the more data can flow through it per second.

The cost of a network connection goes up as bandwidth increases. Thus, a 1 gigabit per second (Gbps) Dedicated Internet Access (DIA) link will be more expensive than one that can handle 250 megabits per second (Mbps) of throughput.

Bandwidth of Signals



The bandwidth of a signal is defined as the difference between the upper and lower frequencies of a signal generated. As seen from the above representation, Bandwidth (B) of the signal is equal to the difference between the higher or upper-frequency (fH) and the lower frequency (fL). It is measured in terms of Hertz (Hz) i.e. the unit of frequency.

Let's understand this better with the help of an example. Whenever you tune into a radio you find various stations at varying particular frequencies. The bandwidth of FM radio is 200 KHz from 88.1 MHz to 101.1 MHz for most places. As you tune, the radio you find various stations at various frequencies.

Neither do all the channel occupy all the frequencies nor is there no channel at every frequency, but every channel has its own unique frequency like an ID.

Given above is a diagram representing various bandwidths of electromagnetic waves and it goes a long way in helping identify the type of wave on basis of its frequency and thus determining its applications. Thus we see that bandwidths exist for all kinds of waves in both the visual and audio spectrums and well beyond them as well.

The audible bandwidth of human ear ranges from 20 Hz to 20,000Hz. Sounds below 20Hz are known as Infrasonic and sounds above 20000Hz are known as ultrasonic. Dogs can hear ultrasonic sounds. Whereas Blue whales can produce infrasonic sounds. So let us now study some applications of various signals in various bandwidths.